Why FDA Withholds Drug Approval: Evidence from 445 Publicly Released Complete Response Letters[1]

Alex Tabarrok
Department of Economics, George Mason University, Fairfax, VA 22030

Revised draft, October 2026

Abstract

Background: The FDA began systematically releasing Complete Response Letters in 2025, providing a broader public record of why drug applications are not approved. This study identifies the most common nonapproval grounds and examines whether they recur across review cycles.

Methods: I analyzed 445 released letters covering 333 applications. A rule-based segmenter isolated substantive deficiencies, and a language model assigned primary and secondary nonapproval grounds. I compared deficiency distributions across periods, examined 112 consecutive-letter pairs, and measured the appearance of another released letter within two years among 250 applications with sufficient follow-up.

Results: Facility inspection was the primary deficiency category in 41.3 percent of letters and 46.2 percent of letters since 2024. Chemistry, manufacturing, and controls (CMC) and clinical efficacy each accounted for 16.4 percent, and clinical safety for 10.1 percent. The primary ground repeated in 58.9 percent of consecutive-letter pairs. Facility inspection remained primary in 42 of 50 pairs that began with it (84.0 percent). Another released letter appeared within two years for 28.6 percent of facility-primary applications and 15.7 percent of applications rejected for other reasons.

Conclusion: Facility inspection is the most common primary deficiency category and frequently recurs in subsequent released letters. Combining facility inspection and CMC also reveals substantial recurrence within manufacturing quality. Incomplete and selective disclosure limits generalization to all FDA nonapproval actions; these results do not establish effects on eventual approval.

Keywords: Food and Drug Administration; Complete Response Letters; drug approval; facility inspections; manufacturing quality.

1. Background

A Complete Response Letter (CRL) communicates an FDA nonapproval action on an application in its present form; it is not necessarily a permanent denial. The letter states, in numbered sections, what must be fixed before the application can be approved. Until recently these letters were confidential. Fragments appeared in approval packages for drugs that eventually got through, which made the public record a record of survivors (Lurie et al., 2015).

In July 2025 the agency released more than 200 letters associated with applications that were later approved. In September it added 89 letters for pending or withdrawn applications and announced that it would release new letters promptly (U.S. Food and Drug Administration, 2025a,b). That release makes a basic descriptive question answerable for the first time. What does the FDA say when it turns a drug down?

Earlier studies answered narrower versions of this question. Sacks et al. (2014) used internal FDA records to study first-time applications for new molecular entities submitted to the Center for Drug Evaluation

[1]The author used ChatGPT Sol extensively in collating and analyzing the data.

and Research (CDER) from 2000 through 2012. That study placed inspection findings inside chemistry, manufacturing, and controls (CMC) and counted CMC as the reason for nonapproval only when safety and efficacy were otherwise satisfactory. Lurie et al. (2015) analyzed 61 nonpublic letters issued from 2008 through 2013 and showed that sponsor announcements omitted most of the FDA's concerns. More recent work also finds manufacturing problems in selected samples. Dilek et al. (2026) reports facility deficiencies in 65 percent of 43 novel therapeutics that were approved from 2020 through 2024 after receiving a letter, while McCarthy and O'Boyle (2026) finds that CMC issues were the most common deficiencies among first-time small-molecule applications submitted from 2013 through 2023.

This paper asks a different question with a broader public corpus. It includes released letters for pending and withdrawn applications, treats facility inspection as distinct from CMC, identifies the primary primary deficiency in each letter, and tracks whether that reason appears again in a later letter. The distinction matters: a failed facility inspection is a compliance problem at a plant, while a CMC deficiency can concern the product or process described in the application.

I downloaded 458 records from the openFDA transparency endpoint on August 28, 2026. The newest letter in that pull is dated August 7, 2026. The 458 records cover 336 applications and 315 companies. Of those records, 445 are Complete Response Letters proper, covering 333 applications and 311 companies from 2002 through the middle of 2026. Each letter runs about eight thousand characters. I code every strict letter by the ground on which the application failed.

The most common ground is facility inspection. An inspector went to a manufacturing site, found conditions that violated good manufacturing practice, and FDA withheld approval. Facility inspection is primary in 41.3 percent of the letters. Chemistry and manufacturing controls, which cover such matters as formulation and process specifications, are a separate category and account for 16.4 percent. Clinical efficacy also accounts for 16.4 percent, and clinical safety for 10.1 percent.

A second pattern shows up among applications with more than one released letter. Eighty-two applications in the corpus have multiple letters, which lets me compare consecutive observed letters. Facility inspection remains the primary reason in 42 of 50 consecutive pairs that begin with it, or 84.0 percent. In a cohort with at least two years of follow-up, another released letter appears within two years for 28.6 percent of facility-primary applications, compared with 15.7 percent of applications rejected for all other reasons. Facility problems are unusually likely to appear in repeat nonapproval letters in the released corpus.

Both results carry limits that Sections 2.1 and 4.1 set out. The corpus has no coverage denominator, matching to Drugs@FDA is selected on approval, and facility identity is redacted in half the relevant letters. This is a first pass at a source that has been public for about a year.

2. Methods

2.1 The data

The openFDA endpoint returns structured records. Each carries the letter date, the sponsor name, the application number, the reviewing division, and the full text of the letter as extracted from a scanned PDF. Bulk PDF archives exist separately and I did not need them.

Coverage is uneven across time. The corpus holds 28 letters from 2002 through 2014, then 121 from 2015 through 2019, 138 from 2020 through 2023, and 158 from 2024 through the August 2026 cutoff. The tilt toward recent years follows from how the release worked. FDA committed to posting new letters close to the time it issues them, and it released a batch of 89 previously unpublished letters from 2024 onward in September 2025. Older material entered the corpus through a different and less systematic route.

I could not calculate what share of all Complete Response Letters the dataset covers because FDA does not publish annual counts of these actions. The 89 letters are treated separately because FDA released them

together as an initial transparency batch in September 2025, rather than through the less systematic process that produced the older records. This makes the recent period more internally consistent, but the batch still does not represent all letters issued during any defined period. FDA's Prescription Drug User Fee Act (PDUFA) reports track application cohorts by fiscal year rather than the number of actions taken each calendar year, so they do not supply the missing denominator. I therefore describe the 2024-forward sample as a distinct but incomplete release cohort.

Drugs@FDA provides additional information, but only for drugs that were eventually approved or tentatively approved. The merge matches 334 of 457 letters with valid application numbers, or 73.1 percent, but only 28.8 percent of letters from 2024 onward. Recent rejected applications are often missing because they have not subsequently been approved. Whether a letter matches therefore depends on the outcome being studied, not on random data availability. For that reason, this paper does not analyze approval outcomes or time to approval.

The matching procedure and its coverage are described in Appendix A.10. The primary estimates use all strict CRLs, regardless of whether an application matches Drugs@FDA.

The study includes letters whose source letter type is exactly COMPLETE RESPONSE. New Drug Applications (NDAs) and Biologics License Applications (BLAs) are included; supplements and combination products are not excluded on that basis. Each source record contributes once. For a letter referring to multiple applications, the first distinct normalized identifier in source order is the primary application for grouping; all associated identifiers are retained. Appendix A.1 details submission references and the exclusion accounting.

Figure 1 reconciles the letter-, application-, and pair-level samples.

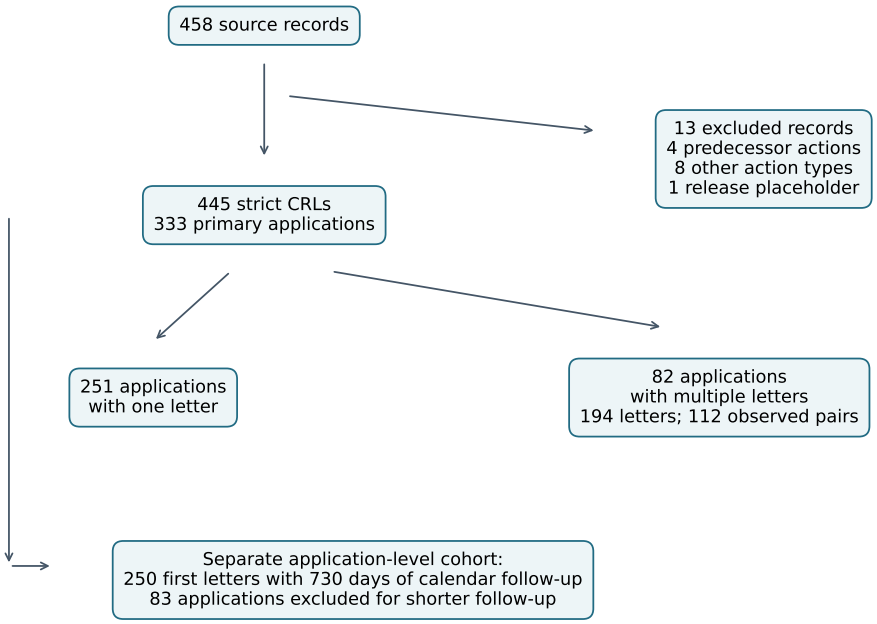


Figure 1: Sample construction. Units are source records, letters, applications, and consecutive observed letter pairs, as labeled. The two-year cohort is a separate application-level selection.

2.2 Coding the letters

A Complete Response Letter contains boilerplate about labeling and resubmission. In a first pass using the full text, labeling was incorrectly identified as a reason for nonapproval in 99 percent of letters.

The coding has two distinct steps. First, a rule-based segmenter removes the letter's introductory and standard closing language, leaving the section that explains why the application could not be approved. Second, a language model classifies that section. It assigns one primary reason for nonapproval and any secondary reasons using ten categories: clinical efficacy, clinical safety, trial design, chemistry and manufacturing controls, facility inspection, stability, bioequivalence, nonclinical toxicology, labeling, and administrative deficiencies.

The original codebook defines the primary category as the ground most directly responsible for the action, judged from FDA's decision language and required remedy, rather than the first heading or longest section. All other substantive grounds are secondary. Facility inspection covers inspection findings, current good manufacturing practice (CGMP) compliance, or facility acceptability; Form FDA 483 records inspectional observations. CMC covers deficient product- or process-quality information. Both categories are assigned only when a distinct application/product-quality deficiency accompanies the facility issue. The original scheme requires one primary category and supplies no mechanical tie-breaker for otherwise coequal deficiencies; the stored rationale records the model's judgment. Appendix A.3 gives examples and the overlap analysis tests sensitivity to this boundary.

I first defined four mutually exclusive inspection categories: an inspection occurred and produced adverse findings, FDA could not complete the inspection, the inspection was pending or conditional, or the outcome was unclear. I also defined a separate flag for contract manufacturers or suppliers. I then used an LLM to apply this predefined scheme to the inspection language extracted from each facility-coded letter. Finally, I tested whether standard inspection language was being mistaken for an actual reason for nonapproval.

I checked each step separately. To check the segmenter, I used an LLM-assisted review of 30 letters to compare the extracted section with the full letter. To check the classification, I used a second LLM review that was blinded to the original model output. The 50-letter review sample was selected before the main classification was run. The methodological justification and detailed comparisons are in Appendix A.5.

Redaction limits the coding in 48 letters and OCR damage limits it in 26. Section 3.5 takes up both.

2.3 Statistical analysis

Category distributions use letters as the unit; transitions use consecutive observed pairs ordered by letter date and then letter identifier within the primary application; two-year outcomes use the first observed strict CRL per application. Eligibility requires at least 730 days between that letter and the August 28, 2026 source-acquisition cutoff. The latest letter in the snapshot is dated August 7; calendar eligibility does not imply complete observation of subsequent FDA actions. A missing second released letter can reflect approval, withdrawal, no resubmission, or an unobserved CRL.

Intervals for single proportions use the Wilson score method with $z = 1.96$. Differences between independent application-level proportions use Newcombe's unpooled Wilson-score method without continuity correction. For proportions p_1, p_0 with Wilson bounds $(L_1, U_1), (L_0, U_0)$, the difference $d = p_1 - p_0$ has bounds $d - \sqrt{(p_1 - L_1)^2 + (U_0 - p_0)^2}$ and $d + \sqrt{(U_1 - p_1)^2 + (p_0 - L_0)^2}$. These are reference intervals conditional on the observed corpus; they do not quantify selective release or coding error. Letter-level Wilson intervals do not adjust for repeated letters within applications. All-pair recurrence is reported descriptively; a first-pair-per-application sensitivity avoids multiple pair contributions. Appendix A.11 also reports a sensitivity excluding same-date pairs.

Manufacturing quality combines facility inspection and CMC; other categories, including stability, retain their original definitions. This sensitivity changes the grouping, not the underlying coding.

3. Results

3.1 Validation results

The segmentation check found no omissions among 30 reviewed letters. The two LLM passes agreed on the primary category in 45 of 50 letters (90 percent; kappa 0.862). The existing blinded human comparison agreed with the main LLM in 41 of 50 letters (82 percent; kappa 0.763). Appendix A.5 reports disagreements and distinguishes independent comparisons from subsequent author review.

3.2 What the letters say

Table 1 gives the primary category distribution for all 445 letters and for the 158 letters dated 2024 or later.

Table 1. Primary nonapproval grounds (letter-level)

Primary ground	Full N	Full share	2024+ N	2024+ share
Facility inspection	184	41.3%	73	46.2%
Chemistry and manufacturing controls	73	16.4%	14	8.9%
Clinical efficacy	73	16.4%	34	21.5%
Clinical safety	45	10.1%	12	7.6%
Nonclinical toxicology	27	6.1%	6	3.8%
Bioequivalence	18	4.0%	7	4.4%
Trial design and conduct	10	2.2%	4	2.5%
Administrative	9	2.0%	5	3.2%
Labeling	5	1.1%	3	1.9%
Stability	1	0.2%	0	0.0%

The 95 percent confidence interval on the full-sample facility share runs from 36.9 to 46.0 percent, and on the recent share from 38.6 to 54.0 percent. These intervals describe the precision of the proportions within the released corpus; they do not correct for incomplete coverage.

Facility inspection is more often primary in BLA letters than NDA letters. It is the primary deficiency in 55.6 percent of BLA letters, compared with 36.3 percent of NDA letters. Chemistry and manufacturing controls account for about 16 percent of both groups. Nonclinical deficiencies appear in 8.2 percent of NDA letters and in no BLA letters.

The mix of nonapproval grounds changes substantially across periods. Facility inspection is the primary deficiency in 10.7 percent of the 28 letters from 2002 through 2014. Its share then rises to 36.4 percent, 46.4 percent, and 46.2 percent in the three later periods. Clinical efficacy follows the opposite pattern at first, falling from 35.7 percent to 9.4 percent before rising to 21.5 percent in the most recent period. These figures describe changes within the published letters, not necessarily changes across all Complete Response Letters. The early groups are small, and FDA released the older and more recent letters through different processes.

The size of the facility share is consistent with a separate FDA analysis based on internal submission data. FDA examined nearly 10,000 original applications and supplements received in fiscal years 2024 and 2025 that required a facility assessment. Among submissions that received a Complete Response Letter, 43 percent had a facility withhold (U.S. Food and Drug Administration, Office of Pharmaceutical Quality, 2026). The figures are not directly comparable: the FDA analysis includes generic applications and supplements and begins with submissions requiring a facility assessment, while this paper covers released NDA and BLA

letters. It provides qualitative context for the importance of facility issues, rather than external validation of the 41.3 percent estimate.

Figure 2 displays the primary distribution. Appendix A.11 reports secondary categories and temporal distributions stratified by NDA versus BLA. Facility inspection and CMC co-occur in 74 of 445 letters (16.6 percent), after incorporating the three documented secondary-code corrections. Combining them as a primary manufacturing-quality category gives 257 of 445 letters (57.8 percent), and 87 of 158 letters from 2024 onward (55.1 percent).

Considering both primary and secondary categories, 259 of 445 letters (58.2 percent) contain facility or CMC findings without a coded clinical deficiency, broadly defined to include efficacy, safety, trial design, and bioequivalence (Appendix Table A6).

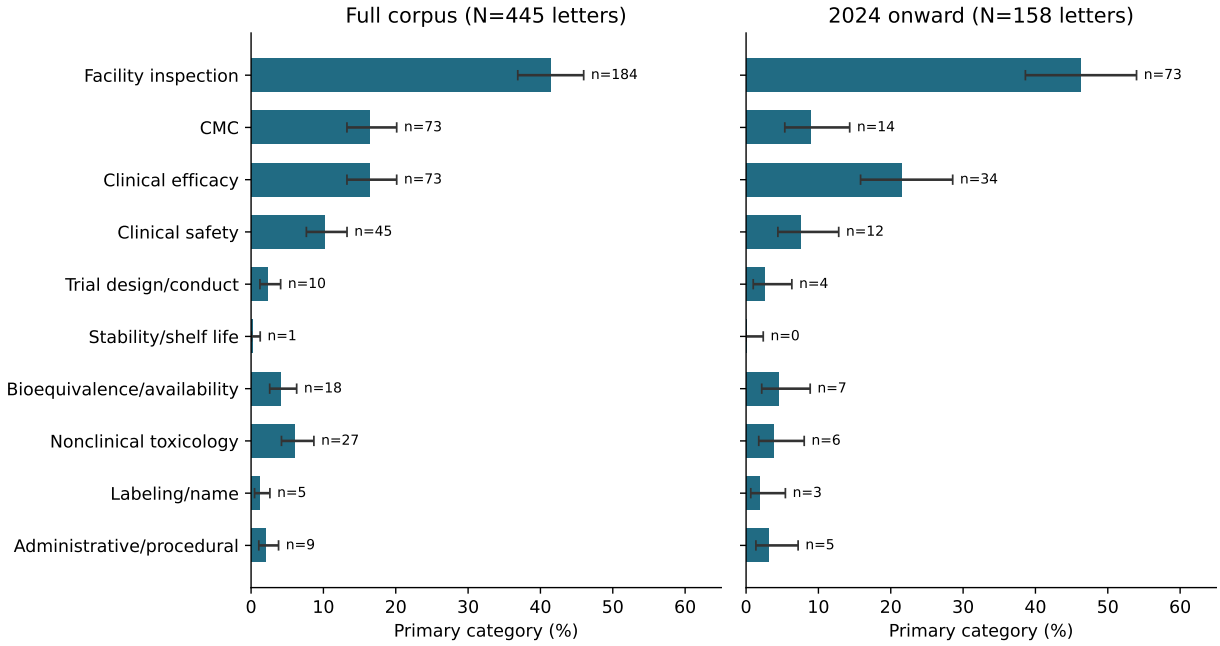


Figure 2: Primary deficiency categories, letter-level. Bars show proportions, labels show counts, and whiskers show 95 percent Wilson intervals. The recent cohort is a subset of the full corpus.

3.3 Inside the facility category

Facility inspection is the primary category in 41.3 percent of the letters, so I check whether routine inspection language inflates this result. Complete Response Letters often mention inspection status even when an inspection did not prevent approval. If the classifier treated those standard statements as nonapproval grounds, the headline estimate would overstate the importance of facility problems. This section includes every letter with facility inspection as either a primary or a secondary category; after the segmenter correction, there are 260 such letters.

Of the 272 letters that mention inspection somewhere in the raw text, five contain no inspection language in the extracted deficiency section. Four of those five contain no substantive facility issue once the standard language is removed. A separate LLM review found no case in which a facility classification rested only on standard language. The 41.3 percent is not a boilerplate artifact.

One possible explanation for the large facility share is that FDA lacked the capacity to complete required inspections, especially at foreign plants and during the pandemic. If that were the main problem, many letters would say that approval was delayed because an inspection could not be completed. The letters do not support that explanation. FDA was unable to complete an inspection in 20 of 260 facility-coded letters, or 7.7 percent, and only two such cases appear from 2024 onward, representing 2.2 percent of facility-coded letters in that period. By contrast, 239 of 260 facility-coded letters, or 91.9 percent, describe an inspection that occurred and found problems. No pending-only case was observed after segmentation; late material can occasionally be missed, so this is not a claim that such cases never occur. Separately, twenty-seven letters mention a contract manufacturer or supplier.

What the letters describe is an inspector who went to a plant and found something wrong.

3.4 Category recurrence in observed letters

Eighty-two applications have multiple released Complete Response Letters in the corpus. Because some have three or more, these applications produce 112 pairs of consecutive letters. Both letters in each pair are coded, allowing me to test whether FDA cited the same primary reason in consecutive observed letters.

The second letter gives the same primary reason as the first in 66 of 112 pairs, or 58.9 percent. The reason repeats in 42 of 50 pairs that begin with facility inspection, 8 of 14 that begin with nonclinical deficiencies, 8 of 15 that begin with clinical efficacy, 3 of 15 that begin with chemistry and manufacturing controls (CMC), and 2 of 8 that begin with clinical safety. All 42 repeated facility cases involve adverse inspection findings in both letters. No pair involves two consecutive nonapproval letters because FDA could not complete an inspection.

The difference between the CMC and facility results prompted a closer look at the underlying letters. In 9 of the 15 pairs that begin with a CMC-primary letter, the second letter is facility-primary. An LLM-assisted review of these nine pairs found that five concern the same site, process step, or quality attribute in both letters. Two involve different problems, and two cannot be compared because the relevant section of the first letter is redacted. In four of the five same-problem pairs, the first letter already included facility inspection as a secondary category. CMC and facility inspection can therefore describe different parts of the same underlying problem, but a movement between them is not automatically a repeat of the same category. I keep them separate in the main analysis and combine them in a sensitivity analysis. The full transition matrix is reported in Appendix A.11.

The median gap between consecutive letters is 359 days when the primary reason repeats and 520 days when it changes. Pairs in which facility inspection remains primary are 169 to 1,256 days apart, with a median of 359 days. These are intervals between letters, not resubmission times; exact resubmission dates are not observed consistently.

These figures include only applications with another released Complete Response Letter in the corpus. They show what happens among observed repeat nonapproval letters, not how often a facility problem remains unresolved across all applications.

The preceding analysis includes only applications with an observed second Complete Response Letter. To avoid conditioning on that observed outcome, I next include every application whose first letter occurred at least two years before the follow-up cutoff. Among these 250 applications, another released letter appears in the corpus within two years for 24 of 84 with a facility-primary first letter, or 28.6 percent. The corresponding rate is 26 of 166, or 15.7 percent, for applications whose first letter cited another primary ground. The difference is 12.9 percentage points, with a 95 percent confidence interval from 2.3 to 24.4 points. Among CMC-primary applications, another released letter appears for 8 of 54, or 14.8 percent. The outcome here is the appearance of any second released Complete Response Letter in the corpus, regardless of the deficiency

cited in it. Because applications must have two full years of follow-up, this analysis largely excludes recent first letters.

The combined-category analysis shows that recurrence extends across the facility–CMC boundary. Manufacturing quality remains primary in 60 of 65 pairs beginning with either category (92.3 percent). Using only the first pair per application, facility inspection repeats in 28 of 33 pairs (84.8 percent), and manufacturing quality in 41 of 45 (91.1 percent). Thus, the recurrence finding is not driven by applications contributing several pairs, but it is not unique to the narrow facility category.

In the two-year application cohort, another released letter is observed for 32 of 138 manufacturing-quality-primary applications (23.2 percent), versus 18 of 112 other applications (16.1 percent). The difference is 7.1 percentage points (95 percent interval, -3.0 to 16.7). Category recurrence among observed pairs and observation of another letter are distinct outcomes; neither measures time to resolving the original deficiency.

3.5 Robustness

Redaction could bias the facility estimate because many letters omit confidential details about manufacturing sites. If the classifier assigned facility inspection when the underlying deficiency was redacted, the reported facility share would be too high.

The data do not show that pattern. Facility inspection is primary in 40.7 percent of letters without classification flags, compared with 41.3 percent of all letters. Among letters whose classification is uncertain because of redaction, the share is lower, at 31.3 percent. From 2024 onward, the facility share is 50.8 percent among letters without flags and 46.2 percent overall. Excluding redaction-uncertain letters would therefore raise, not lower, the estimated facility share.

OCR-uncertain letters run the other way: facility inspection is primary in 73.1 percent, compared with 40.7 percent among letters without classification flags. OCR problems are especially common in facility letters because plant names, addresses, inspection details, and observations often scan poorly. The standard language stating that an inspection found deficiencies and that the problems must be resolved before approval usually remains readable.[2]

Facility inspection accounts for 41.3 percent of all letters, 40.7 percent of letters without classification flags, and 39.4 percent after excluding every OCR-uncertain letter. From 2024 onward, the corresponding shares are 46.2, 50.8, and 44.9 percent. The conclusion that facility inspection is the most common reason for nonapproval does not depend on how the OCR-uncertain letters are treated.

4. Discussion

The letters cannot show why facility problems recur. Some plants may have difficulty correcting the conditions found during an inspection. In other cases, the problem may be the time and work required to demonstrate that the corrections are adequate. Distinguishing these explanations would require inspection records, including the underlying Form 483 observations, subsequent inspection classifications, and evidence of how and when each problem was resolved.

Related work reaches a similar conclusion at a later stage of the product life cycle. Manufacturing-quality problems are a major source of drug shortages, and weak market rewards for quality can reduce investment in reliable production (Woodcock and Wosinska, 2013; U.S. Food and Drug Administration, Drug Shortages Task Force, 2019). Those studies concern the supply of drugs already on the market, whereas this paper concerns entry. Together, they show that facility quality can constrain both approval and continued availability.

[2]A second LLM reviewed all 26 OCR-uncertain letters without access to the original classifications. It confirmed all 19 facility assignments and agreed with 25 of the 26 classifications overall. The remaining letter did not contain enough readable text to classify.

At the stage represented by these letters, clinical efficacy and safety are less often the primary obstacle than facility inspection. These findings suggest that manufacturing-site readiness is not a peripheral compliance issue but a central determinant of whether FDA can approve an application. Sponsors may therefore benefit from treating inspection readiness, including oversight of contract facilities, as part of development strategy rather than as a late-stage operational task. More consistent disclosure of facility identity and deficiency type would allow researchers and policymakers to distinguish sponsor-controlled problems from failures in outsourced production.

4.1 Limitations

The released letters do not show whether recurring facility problems are more common at sponsors' own plants or at contract manufacturers. FDA often removes information identifying the plant. I examined ownership in the 257 letters initially classified as facility cases, before correcting the method used to separate deficiencies from standard letter text. Ownership was unclear in 165 letters. In 128 of these, the plant's identity had been removed—49.8 percent of all 257 letters. Sponsor-owned plants could be identified in 65 letters and third-party plants in 27.

Missing ownership information also limits comparisons across review cycles. Among the 42 pairs of letters in which facility inspection remained the primary deficiency category, ownership could be identified in 19. Among the eight pairs in which the primary reason changed, ownership could be identified in only one. These gaps prevent a reliable comparison of sponsor-owned plants and contract manufacturers. Disclosing plant identities more often would make that comparison possible.

The study has three other limits. First, FDA does not publish the total number of Complete Response Letters issued, so I cannot determine what share is included here. Differences across periods in Section 3.2 may therefore reflect which letters were released. Second, Drugs@FDA mainly covers applications that were eventually approved or tentatively approved. Restricting the analysis to applications found there would leave out many applications that have not been approved. Third, resubmission dates can be recovered only when a later rejection letter mentions them. This leaves out resubmissions that succeeded, so I do not analyze resubmission timing.

5. Conclusion

First, facility inspection is the most common reason given for nonapproval, accounting for 41.3 percent of the released letters and remaining between 39.4 and 41.3 percent under alternative treatments of uncertain classifications. This finding is not a boilerplate artifact and primarily reflects inspections that occurred and found problems.

Second, facility inspection problems are unusually likely to appear in repeat nonapproval letters in the released corpus. Among applications with an observed second letter, facility inspection remained the primary reason in 84.0 percent of cases that began with it. Among all applications with two full years of follow-up, another released letter appears in the corpus for 28.6 percent of those with a facility-primary first letter, compared with 15.7 percent of applications rejected for other reasons. CMC-primary applications have a rate of 14.8 percent. The combined analysis also shows substantial recurrence within manufacturing quality, so recurrence is not unique to facility inspection. These results do not establish whether facility problems reduce the probability of eventual approval.

Data availability statement

The complete study data and code are available as a replication ZIP at alextabarrok.com. The archive includes frozen FDA downloads, extracted text, stored model outputs, existing validation materials, linkage files,

analysis datasets, and reproduction instructions. Results can be rebuilt offline from stored classifications or independently recoded using the supplied text and prompts. Fresh model calls may yield different classifications. The source data were downloaded on August 28, 2026; the latest letter is dated August 7, 2026.

Author Contributions

Alex Tabarrok is the sole author and conceived and designed the study and interpreted the data. Tabarrok thanks ChatGPT Astra and Claude Fable for error checking, productive comments, assistance with analysis, and editorial summarizing.

References

Sophie Dilek, Richard H. Woods, Jeromie Ballreich, Thomas J. Moore, and G. Caleb Alexander. Deficiencies delaying prescription drug approvals by the U.S. Food and Drug Administration, 2020–2024. *Therapeutic Innovation & Regulatory Science*, 60(3):837–846, 2026. doi: 10.1007/s43441-026-00921-3.

Fabrizio Gilardi, Meysam Alizadeh, and Mael Kubli. ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, 120(30):e2305016120, 2023. doi: 10.1073/pnas.2305016120.

Seungone Kim, Dongkeun Yoon, Kiril Gashteovski, et al. On the limits and opportunities of AI reviewers: Reviewing the reviews of Nature-family papers with 45 expert scientists. arXiv:2605.20668, 2026. Preprint, May 20.

Peter Lurie, Harpreet Singh Chahal, Daniel W. Sigelman, Sylvie Stacy, and Joshua Sclar. Comparison of content of FDA letters not approving applications for new drugs and associated public announcements from sponsors: Cross sectional study. *BMJ*, 350:h2758, 2015. doi: 10.1136/bmj.h2758.

Gemma McCarthy and Niamh M. O'Boyle. Analysis of FDA approvals, delays and denials of new drug applications, 2013–2023. *Drug Discovery Today*, 31(4):104739, 2026. doi: 10.1016/j.drudis.2026.104739.

Leonard V. Sacks, Hala H. Shamsuddin, Yuliya I. Yasinskaya, Khaled Bouri, Michael L. Lanthier, and Rachel E. Sherman. Scientific and regulatory reasons for delay and denial of FDA approval of initial applications for new drugs, 2000–2012. *JAMA*, 311(4):378–384, 2014. doi: 10.1001/jama.2013.282542.

Petter Törnberg. Large language models outperform expert coders and supervised classifiers at annotating political social media messages. *Social Science Computer Review*, 43(6), 2025. doi: 10.1177/08944393241286471.

U.S. Food and Drug Administration. FDA embraces radical transparency by publishing complete response letters. FDA press announcement, July 2025a. July 10, 2025.

U.S. Food and Drug Administration. FDA announces real-time release of complete response letters, posts previously unpublished batch of 89. FDA press announcement, September 2025b. September 4, 2025.

U.S. Food and Drug Administration, Drug Shortages Task Force. Drug shortages: Root causes and potential solutions. Technical report, U.S. Food and Drug Administration, 2019. Updated February 2020; available from FDA.

U.S. Food and Drug Administration, Office of Pharmaceutical Quality. Report on the state of pharmaceutical quality, FY2025. Technical report, U.S. Food and Drug Administration, 2026. Available from FDA.

Janet Woodcock and Marta Wosinska. Economic and technological drivers of generic sterile injectable drug shortages. *Clinical Pharmacology & Therapeutics*, 93(2):170–176, 2013. doi: 10.1038/clpt.2012.220.

Appendix: Data, Code and Methods

All letter data use the openFDA transparency endpoint at data vintage 2026-08-07. The 2026-08-28 follow-up cutoff is used only to determine whether two years have elapsed since an application's first observed letter. Classification code, blinded validation samples, segmenter rules, and the analysis panel are included with all frozen source data in the accompanying complete replication archive. No result in this paper requires the bulk PDF archives.

All excerpts below come from the released letters themselves. FDA correspondence is a work of the United States government and carries no copyright. Text is reproduced as it appears in the openFDA extraction, including scanning errors, so that readers can see what the classifier had to work with. Redaction markers appear in the source as `(b) (4)`, the FOIA exemption for confidential commercial information.

A.1 Corpus

I downloaded 458 records from the openFDA transparency endpoint at `https://api.fda.gov/transparency/crl.json` on August 28, 2026. The newest letter is dated August 7, 2026. Each record carries the letter date, sponsor name, application number, reviewing division, approver, and the full text of the letter as extracted from a scanned PDF.

Of the 458 records, 445 are Complete Response Letters proper and are used for every headline figure. Four predecessor not-approvable actions have no source letter type and are classified but excluded from the strict-CRL tables. The other nine exclusions are eight other action types (four tentative approvals, one refusal to file, one rescission, one provisional determination, and one corrected provisional determination) and one release placeholder with no usable application identifier. These are action-type exclusions, not nine structural failures.

The 445 strict letters cover 333 unique primary applications and 311 companies. Median letter length is about 8,700 characters.

The source inclusion rule does not exclude supplements or combination products. Four strict CRLs have explicit parsed supplement references; thirteen have explicit original-submission references and 428 have no parsed submission reference, so absence of a supplement reference is not proof of an original submission. One letter references two original submissions; one strict letter references multiple distinct applications and is counted once using the first source-listed normalized application. The linkage file preserves all associated applications. Combination-product letters follow the same rule and are coded for their substantive deficiencies.

The original ordering is by letter date and then letter identifier. Two applications have two records on the same date; these produce two zero-day pairs. They are retained in the original 112-pair definition but excluded in a reported sensitivity. They must not be interpreted as two distinct dated review cycles. The 82 multiple-letter applications contribute 194 letters, hence $194 - 82 = 112$ pairs; the remaining 251 applications contribute one letter each.

A.2 Segmentation

A letter contains an opening paragraph, a body of numbered deficiencies, and closing sections on labeling, resubmission procedure, and dispute resolution. Only the body carries the rejection ground. Feeding whole letters to a classifier produces the labeling artifact described in Section 2.2 of the paper.

The segmenter selects a structural start and truncates at the earliest structural end. Start cues use recurring phrasing that introduces the deficiency list, most often a sentence naming the reasons for the action or announcing that deficiencies follow. When no cue matches, the segmenter falls back to the decision sentence and then to the first section heading.

End patterns, in the order the segmenter applies them:

```
^OTHER REQUEST(S) AND COMMENT(S)$
^SAFETY UPDATE$
^ADDITIONAL COMMENT(S)$        (conditional; see below)
^Within [number] days after [the] date of this letter
^Within one year after [the] date of this letter
^A resubmission must fully address
^If you decide to resubmit
^You are required to resubmit
```

The late fallback expressions are `^You may request a meeting`, `^If you have any questions`, and `^Sincerely`.

Three of these headings are conditional rather than terminal. The segmenter scans forward past `SAFETY UPDATE`, `OTHER REQUESTS AND COMMENTS`, and `ADDITIONAL COMMENTS` and continues the segment when approval-binding language follows.

That lookahead was added after the fact. The original rule treated `SAFETY UPDATE` as always terminal, and four letters place inspection material after it. Three lost the material entirely, which suppressed three facility codes and three cases of uncompleted inspection. Re-running the patched segmenter over all 449 classified actions recovers exactly those three and nothing else. No other end pattern shows the vulnerability. The frozen panel was not rebuilt; corrections are carried as a sensitivity row and stored in `segmentation_corrections_v20260807.csv`.

An LLM-assisted audit of the patched segments for 30 letters, covering four pre-2008 letters, thirteen from 2008 through 2023, and thirteen from 2024 forward, found no cases in which the segmenter omitted a stated reason for withholding approval. Three letters raised automatic flags before review and all three were confirmed correct. No letter produced an empty deficiency segment.

A.2.1 What the segmenter removes

This is the labeling boilerplate that appears in nearly every letter, from NDA 210649 (AB Pharmaceuticals, September 12, 2018):

> PRESCRIBING INFORMATION
>
> 2. We reserve comment on the proposed labeling until the application is otherwise adequate. We encourage you to review the labeling review resources on the PLR Requirements for Prescribing Information and Pregnancy and Lactation Labeling Final Rule websites, including regulations and related guidance documents and the Selected Requirements for Prescribing Information (SRPI) — a checklist of important format items from labeling regulations and guidances.

Salamandra's letter for NDA 208647, issued two years earlier, contains the same paragraph with the same wording. A keyword counter reads both as labeling deficiencies. Neither is one.

The resubmission language is equally standard, from the same AB Pharmaceuticals letter:

> Within one year after the date of this letter, you are required to resubmit or take other actions available under 21 CFR 314.110. If you do not take one of these actions, we may consider your lack of response a request to withdraw the application under 21 CFR 314.65.

A.3 Category scheme

The main LLM classification assigns each segmented letter one primary category and any number of secondary categories:

Code	Category	Scope
1	Clinical efficacy	Evidence does not establish effectiveness
2	Clinical safety	Specific safety signal or unfavorable risk-benefit
3	Trial design and conduct	New or different trial required, data integrity, GCP
4	Chemistry and manufacturing controls	Formulation, specification, process validation
5	Facility inspection	Inspection findings, CGMP status, site compliance
6	Stability and shelf life	
7	Bioequivalence and bioavailability	
8	Nonclinical toxicology	
9	Labeling and proprietary name	Substantive, beyond reserved-comment boilerplate
10	Administrative and procedural	Incomplete submission, missing data, REMS

Category 5 is subdivided:

Code	Subcategory	Patched N (of 260 facility-coded)
5a	Inspection occurred, adverse findings	239
5b	Inspection could not be completed	20
5c	Pending or conditional statement only	0
5e	Unclear	1
5d	Third-party site overlay (flag, not exclusive)	27

The original frozen panel contained 257 facility-coded letters and 17 cases in 5b. The patched figures above add the three recovered 5b cases. They change no primary category and no 2024-forward result.

The classifier also records whether a new clinical trial is explicitly required, whether the deficiency lies with a contract manufacturer, and whether the letter states the application is otherwise approvable.

A.4 Model configuration

Each letter received one call from the main model, using only the segmented text. The prompt supplies the category definitions above, requires structured JSON output, and requires a rationale naming the approval-binding language the classifier relied on. Rationales are stored and are what made several of the checks in this appendix possible. The stored run records identify model alias `gpt-5.6-terra`, low reasoning effort, and calls made on August 28, 2026. The request code does not set temperature. Stored responses retain model identifiers, timestamps, prompt and segment hashes, and response IDs. The validation reviews and sensitivity checks were separate from the main classification run.

The structured schema requires one of the ten primary categories and permits a `failed` classification-status flag; it does not permit a `cannot_determine` primary category. Two classified actions are flagged failed. The original one-primary coding is retained, with the existing uncertain-text sensitivities reported separately.

A.5 Validation

A 50-letter random sample was drawn before the main classification ran. A second LLM classified those letters without access to the main LLM output. Twenty-one of the 50 are dated 2024 or later. Agreement between the two LLM classifications on the primary category is 45 of 50, or 90 percent (Cohen's kappa 0.862).

A later independent human validation used the same blinded 50-letter packet. The human reviewer selected exactly one primary category per letter. No secondary categories, rationales, trial-requirement indicators, manufacturer-ownership fields, approval-status fields, or text-quality flags were collected from the human reviewer. Agreement between the human primary categories and the main LLM is 41 of 50, or 82 percent (Cohen's kappa 0.763).

Comparison	Agreements	Agreement	Cohen's kappa
Main LLM versus blinded second LLM	45/50	90%	0.862
Main LLM versus blinded human primary-only review	41/50	82%	0.763

The category-level table below describes the blinded LLM-to-LLM comparison.

Category	Second N	Agreed/second	Main N	Agreed/main
Clinical efficacy	5	5/5	6	5/6
Clinical safety	4	4/4	6	4/6
Trial design and conduct	1	1/1	1	1/1
Chemistry and manufacturing controls	11	10/11	10	10/10
Facility inspection	23	22/23	22	22/22
Bioequivalence	1	0/1	0	—
Nonclinical toxicology	4	3/4	3	3/3
Labeling	1	0/1	1	0/1
Administrative	0	—	1	0/1

Facility inspection carries most of the paper and performs best in the LLM-to-LLM check. The second LLM agrees with all 22 facility assignments made by the main LLM and identifies one additional facility case. The two LLMs also agree on all ten main-LLM chemistry assignments, while the second LLM identifies one additional chemistry case.

The weak cells are labeling, bioequivalence, and administrative, each resting on a single observation. Administrative deserves particular caution. Its one main-LLM assignment in the LLM-to-LLM validation sample is not confirmed. The disagreement is instructive: the letter's binding product-quality section is fully redacted, and the classifier fell back on the procedural text that survived. The administrative recurrence rate rests on only four originating pairs and is not emphasized; the full transition matrix retains those observations for completeness.

The human comparison differs chiefly at the facility boundary: the human reviewer confirms 17 of the main LLM's 22 facility assignments and assigns the other five to CMC, labeling, or clinical safety. Across all nine human-to-main-LLM disagreements, a case-by-case check by the author supports the LLM primary label in six cases, supports the human label in two redaction-sensitive cases, and finds one combination-product case genuinely ambiguous. This author review was post hoc and not independent; it does not establish that the LLM was more accurate or replace blinded human validation. The analysis retains the original model classifications, reports the human comparison transparently, and examines sensitivity to the facility–CMC boundary.

A.5.1 Rationale for model-assisted coding

Human annotation is not automatically a more accurate benchmark. Gilardi et al. (2023) find that ChatGPT outperforms crowd workers on several text-annotation tasks, while Törnberg (2025) finds that GPT-4 outperforms expert human coders on a political-message classification task with observable ground truth. These findings support evaluating coding performance rather than assuming a hierarchy between human and model judgments.

Recent evidence also concerns scientific evaluation. In a May 2026 preprint, Kim et al. (2026) report that 45 domain scientists evaluated 2,960 criticisms of 82 Nature-family papers. A GPT-5.2 reviewing agent scored above each paper's top-rated human reviewer on a composite of correctness, significance, and evidential support (60.0 versus 48.2 percent). The study also identifies shared model weaknesses and concludes that AI reviewers complement human reviewers. This is evidence about scientific critique, not a direct validation of regulatory-letter classification.

The present study uses model coding, the existing independent model and human comparisons, and text-quality sensitivities. Model-to-model agreement measures consistency, not ground-truth accuracy; neither external annotation studies nor the peer-review study establishes accuracy on these FDA letters. The full letter text, codebook, prompts, model outputs, existing human coding workbook, and disagreement records are supplied to allow independent assessment and alternative coding.

A.5.2 The five LLM-to-LLM disagreements

Redaction affects four of the five disagreements.

Application	Date	Second LLM	Main LLM	What happened
BLA 761326	2024-07-10	Facility	Clinical safety	Both grounds present; letter covers two submissions with different binding issues
NDA 217433	2024-07-15	Labeling	Clinical safety	Device human-factors boundary; some product-quality text is redacted
NDA 219357	2024-12-23	CMC	Administrative	Product quality body fully redacted; only category label survives
NDA 209531	2025-09-22	Nonclinical	Labeling	Nonclinical and product quality detail redacted; responsive labeling text remains visible
NDA 220138	2026-06-15	Bioequivalence	Clinical efficacy	PK bridging boundary; CMC and nonclinical details are redacted

Two of the five LLM-to-LLM disagreements are boundary disagreements between defensible readings rather than errors.

A.6 Classification status flags

Every letter carries a status recording whether the coding rests on legible, unredacted text. The following distribution is letter-level (N = 445).

Status	Strict CRLs	Share
`ok`	369	82.9%
`uncertain_redaction`	48	10.8%
`uncertain_ocr`	26	5.8%
`failed`	2	0.4%

Facility share by status: 40.7 percent among `ok`, 31.3 percent among `uncertain_redaction`, 73.1 percent among `uncertain_ocr`.

The OCR association was large enough to warrant review of all 26 `uncertain_ocr` letters. A separate LLM review was blinded to the original output. It agreed with all 19 facility assignments and with 25 of 26 classifications overall. The single disagreement is NDA 219846, where only the headings `NONCLINICAL` and `PRODUCT QUALITY` survive scanning and the second review declines to classify. The reason for withholding approval was legible in 24 of 26 letters.

The mechanism is a property of the documents. A facility deficiency names a plant, an address, an inspection date, and a set of observations, and that material scans badly. The remedy sentence is standardized and survives intact.

A.7 Boilerplate accounting for the facility category

Because facility inspection carries the paper's central result, the category was checked against the possibility that standard inspection language was being promoted to primary status.

Measure	Full sample	2024 forward
Letters mentioning inspection in raw text	272	96
Entire literal mention removed by segmentation	5	0
Removed with no substantive facility content left	4	0
Confirmed pure template	0	0
Facility assigned among confirmed pure template	0	0

We separately reviewed all five letters identified only in the unsegmented text. Three contained genuine uncompleted-inspection conditions that had been omitted because of the `SAFETY UPDATE` segmentation error; these cases prompted the correction described in Appendix A.2. None was classified as a facility case solely because of template language. The zero reported in row 5c therefore means that no additional cases were observed after segmentation, not that no such cases exist in the full population, because segmentation may occasionally omit material appearing late in a letter.

A.8 Worked examples

A.8.1 A failed inspection (5a), NDA 205920, Armstrong Pharmaceuticals, May 22, 2014

> During a recent inspection of the manufacturing facility for this application, our field investigator conveyed current Good Manufacturing Practices (cGMP) deficiencies to the representative of the facility. [...] Satisfactory resolution of these cGMP deficiencies is required before this application may be approved.

Primary: facility inspection, 5a. In the LLM panel, the separate nonclinical issue about an excipient is recorded as a secondary category. This is the modal shape of a facility letter. An inspection happened, findings were conveyed on a form 483, and approval waits on resolution.

A.8.2 An inspection that could not happen (5b), BLA 761082, Kashiv BioSciences, August 2, 2021

> FDA must assess the ability of that facility to conduct the listed manufacturing operations in compliance with CGMP. Due to U.S. Government and/or Agency-wide restrictions on travel, we were unable to conduct an inspection of the Kashiv Biosciences LLC facility during the current review cycle, and the application cannot be approved until the required FDA inspection is conducted and the findings are assessed with regard to this application.

Primary: facility inspection, 5b. MediWound's letter for BLA 761192, issued two months earlier, uses nearly identical language. Both date from the pandemic travel restrictions, when many of the 20 observed 5b cases occurred. The subcategory runs at 2.2 percent of facility letters in the 2024-forward window.

A.8.3 A chemistry deficiency (category 4), NDA 205394, RedHill Biopharma, January 31, 2014

> The acceptance criterion for the disintegration is overly broad. Tighten the specification for disintegration to a limit that is justified by data and patient requirements.
>
> The acceptance criterion for the dissolution in the drug product specification was revised [...] Update the drug product release and stability specifications to reflect this change and provide dissolution data for the last time point on the stability samples with this method.

Primary: chemistry and manufacturing controls (CMC). Note what distinguishes this from the Armstrong letter. The deficiency concerns a numerical specification rather than inspection findings at a named site. The distinction matters for Section 3.4 of the paper, where CMC-primary applications have another released letter appear less often than facility-primary applications.

A.8.4 An efficacy deficiency (category 1), BL 125827, Replimune, April 10, 2026

> [...] subject matter experts from Oncology Center of Excellence unanimously determined data presented are insufficient to conclude substantial evidence of effectiveness of vusolimogene oderparepvec for treatment of unresectable advanced cutaneous melanoma. [...] The deficiencies for each of the studies are outlined here. For RPL-001-16: Inability to isolate the contribution of vusolimogene oderparepvec when administered in combination with nivolumab

Primary: clinical efficacy. Efficacy letters name the study, the endpoint, and the inferential problem. The two classification passes agree on all five efficacy cases in the validation sample.

A.8.5 Redaction plus scanning damage, NDA 218317, NeuroDerm, May 24, 2024

> (b) (4)
>
> FACILITY INSPECTIONS
>
> (9)
>
> (10
>
> Following pre-approval inspection of om
>
> , listed in the application, FDA conveyed deficiencies to the representative of the facility. The facility should provide satisfactory responses to these deficiencies to the FDA office indicated on the form FDA-483 prior to your complete response to your application.

This one letter shows both problems at once. The `(b) (4)` marker removes preceding material. The facility name has been redacted and what survives the scan is the fragment `om`. The stray `(9)` and `(10` are misread section numbers. The classifier can still see that a pre-approval inspection produced deficiencies conveyed on a form 483, which is enough to assign 5a, and it cannot see whose plant it was.

About half of the 257 letters with ownership coding look like this, which is why Section 4.1 of the paper reports that the sponsor-versus-contractor comparison cannot be made from this corpus.

A.9 Known limits of the classification

The CMC and facility boundary is the weakest part of the scheme. An LLM-assisted review of the nine pairs that move from a CMC-primary first letter to a facility-primary second letter finds five where both letters concern the same underlying site or process problem. Two involve different problems and two cannot be determined because the relevant text is redacted. The main persistence analysis keeps CMC and facility inspection separate. The review explains why some transitions may reflect the boundary between the categories, but it does not recode them as repetitions of the same ground.

Redaction obscured the decisive deficiency in 48 letters, making their classifications uncertain. Four of the five disagreements between the two LLM reviews, and two of the nine disagreements between the human reviewer and the main LLM, involved redaction. Facility deficiencies were less common among redaction-flagged letters than among letters without classification flags, suggesting that redaction does not inflate the paper's headline facility estimate. Its effect on the other categories cannot be determined. Stability, labeling, and administrative deficiencies appeared too rarely in the validation sample to support category-specific conclusions.

Nothing in the scheme captures severity. A letter listing one specification revision and a letter listing a dozen CGMP violations both count once.

A.10 Matching coverage

I examined whether the lower Drugs@FDA match rate for BLAs could hide facility cases. Drugs@FDA includes CDER-regulated therapeutic biologics but generally does not include products reviewed by the Center for Biologics Evaluation and Research (CBER), such as vaccines, blood products, and cell and gene therapies. A CBER-regulated BLA may therefore be absent from Drugs@FDA even if it was eventually approved. This matters because BLA letters are especially likely to cite facility inspection. Facility inspection is primary in 25 of 42 unmatched BLA letters, or 59.5 percent, compared with 40 of 75 matched BLA letters, or 53.3 percent. The difference runs in the direction of modestly understating facility problems in a matched-only sample. CBER explains only part of the lower match rate: of the 33 unmatched BLA applications, 12 were reviewed by CBER, 16 by CDER, and five could not be assigned to a center. The paper's facility estimates use the full letter corpus rather than the Drugs@FDA-matched subset, so this difference does not affect the headline result.

A.11 Overlap, transitions, and sensitivity analyses

Table A1. Primary and secondary categories (letter-level). All 445 strict CRLs are included. Secondary counts exclude the primary assignment. The three documented recovered secondary facility codes are included; no primary category changes. Categories can co-occur, so any-category percentages need not sum to 100.

Category	Primary	Secondary	Any	Any (%)
Facility inspection	184	76	260	58.4
CMC	73	82	155	34.8
Clinical efficacy	73	15	88	19.8
Clinical safety	45	39	84	18.9
Trial design/conduct	10	88	98	22.0
Stability/shelf life	1	65	66	14.8
Bioequivalence/availability	18	26	44	9.9
Nonclinical toxicology	27	25	52	11.7
Labeling/name	5	58	63	14.2
Administrative/procedural	9	22	31	7.0

Table A2. Consecutive observed category transitions (pair-level). Each cell is a count followed by the row percentage in parentheses. Rows index the first letter and columns the next observed letter; N is the row denominator. Fac = facility inspection; Eff = efficacy; Saf = safety; Trial = trial design/conduct; Stab = stability; Bio = bioequivalence/availability; Noncl = nonclinical; Label = labeling/name; Admin = administrative. Dashes denote no originating pairs. The machine-readable table also supplies unrounded row proportions.

First / next	Fac	CMC	Eff	Saf	Trial	Stab	Bio	Noncl	Label	Admin	N
Facility inspection	42 (84)	6 (12)	1 (2)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	1 (2)	50
CMC	9 (60)	3 (20)	0 (0)	1 (7)	1 (7)	1 (7)	0 (0)	0 (0)	0 (0)	0 (0)	15
Clinical efficacy	1 (7)	1 (7)	8 (53)	3 (20)	0 (0)	0 (0)	0 (0)	2 (13)	0 (0)	0 (0)	15
Clinical safety	3 (38)	2 (25)	1 (12)	2 (25)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	8
Trial design/conduct	2 (50)	0 (0)	0 (0)	1 (25)	1 (25)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	4
Stability/shelf life	—	—	—	—	—	—	—	—	—	—	0
Bioequivalence/availability	1 (50)	1 (50)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	2
Nonclinical toxicology	4 (29)	1 (7)	0 (0)	0 (0)	0 (0)	0 (0)	0 (0)	8 (57)	0 (0)	1 (7)	14
Labeling/name	—	—	—	—	—	—	—	—	—	—	0
Administrative/procedural	1 (25)	0 (0)	0 (0)	0 (0)	1 (25)	0 (0)	0 (0)	0 (0)	0 (0)	2 (50)	4

Table A3. Recurrence sensitivities (pair-level). Manufacturing quality merges facility inspection and CMC only. First-pair results use at most one pair from each application. The same-date sensitivity removes two zero-day pairs; neither began with facility inspection or CMC. For first pairs, the Wilson intervals are 46.5–67.5 percent for any same-category recurrence, 69.1–93.3 percent for facility recurrence, and 79.3–96.5 percent for manufacturing-quality recurrence. All-pair estimates are descriptive because pairs within applications are dependent.

Sample / definition	N	Repeated	%
All observed pairs / all categories	112	66	58.9
All observed pairs / Facility inspection	50	42	84.0
All observed pairs / CMC	15	3	20.0
All observed pairs / all categories (merged)	112	81	72.3
All observed pairs / manufacturing quality	65	60	92.3
First pair per application / all categories	82	47	57.3
First pair per application / Facility inspection	33	28	84.8
First pair per application / CMC	12	3	25.0
First pair per application / all categories (merged)	82	57	69.5
First pair per application / manufacturing quality	45	41	91.1
Exclude same-date pairs / all categories	110	64	58.2
Exclude same-date pairs / Facility inspection	50	42	84.0
Exclude same-date pairs / CMC	15	3	20.0
Exclude same-date pairs / all categories (merged)	110	79	71.8
Exclude same-date pairs / manufacturing quality	65	60	92.3

Table A4. Another released CRL within 730 days (application-level). The first observed strict CRL defines the category. All 250 applications have at least 730 days of calendar follow-up to August 28, 2026. Intervals are Wilson intervals. The facility versus other difference is 12.9 percentage points (Newcombe interval 2.3–24.4); the manufacturing-quality versus other difference is 7.1 points (-3.0 to 16.7).

First-letter category	Applications	Another CRL	%	95% CI
Facility	84	24	28.6	20.0–39.0
Other than facility	166	26	15.7	10.9–22.0
Manufacturing quality	138	32	23.2	16.9–30.9
Other than manufacturing quality	112	18	16.1	10.4–24.0

Table A5. Facility-primary letters by period and application type (letter-level). These are within-corpus descriptions; issuance period is not release date. Small early BLA cells should not be interpreted as precise trends.

Period	Type	Letters	Facility	%
2002–2014	NDA	28	3	10.7
2015–2019	BLA	15	3	20.0
2015–2019	NDA	106	41	38.7
2020–2023	BLA	41	24	58.5
2020–2023	NDA	97	40	41.2
2024–2026	BLA	61	38	62.3
2024–2026	NDA	97	35	36.1

Table A6. Manufacturing and clinical findings (letter-level). Presence includes primary and secondary categories, with the three documented secondary-code corrections. Manufacturing means facility inspection or CMC. The broad clinical definition includes efficacy, safety, trial design/conduct, and bioequivalence/bioavailability. Cells report counts and percentages of all letters in the column: 445 overall and 369 with classification status `ok`. The latter excludes redaction, OCR, and failed-classification flags.

Coded findings	All letters	Unflagged
Manufacturing without clinical	259 (58.2)	208 (56.4)
Both	82 (18.4)	72 (19.5)
Clinical without manufacturing	86 (19.3)	76 (20.6)
Neither	18 (4.0)	13 (3.5)

Manufacturing findings without a coded clinical deficiency appear in 259 of 445 letters (58.2 percent), and in 208 of 369 unflagged letters (56.4 percent). Excluding bioequivalence/bioavailability from the clinical definition gives 275 of 445 letters (61.8 percent). These groups can include other findings, such as nonclinical or labeling deficiencies; absence of a coded clinical finding does not establish that all clinical requirements were satisfied. Letter-level indicators and both clinical definitions are supplied in the replication archive.